

Hypothesis H1: Physical Activity Affects Sleep Quality

1. Hypothesis Statement

- **Null Hypothesis (H_0):** The mean sleep quality is the same for physically active and inactive individuals.
 - **Alternative Hypothesis (H_1):** The mean sleep quality is different for physically active and inactive individuals.
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2. Data and Variables

- **Dataset:** Sleep and Lifestyle dataset
 - **Independent Variable:** Physical activity (Active vs Inactive)
 - Threshold: ≥ 7000 daily steps = Active
 - < 7000 steps = Inactive
 - **Dependent Variable:** Quality of Sleep (numeric)
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3. Methodology

1. Data Preparation:

- Selected only relevant columns (Daily Steps, Quality of Sleep)
- Dropped rows with missing values

2. Group Creation:

- Created a binary column Active based on daily steps

3. Variance Check (F-test):

- Calculated variance of sleep quality in each group
- F-statistic = 0.551 $\rightarrow$ close to 1 $\rightarrow$ variances are approximately equal

4. Two-sample t-test (equal variances):

- Calculated t-statistic to test for difference in mean sleep quality
- Degrees of freedom = $n_1 + n_2 - 2$ (large sample)
- Significance level $\alpha = 0.05$

- Decision rule: reject H_0 if $|t_{\text{calculated}}| > t_{\text{critical}}$
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4. Results

- **t-statistic:** 5.923
- **p-value:** 7.2×10^{-9}
- **t-critical (two-tailed, $\alpha = 0.05$, large df):** ≈ 1.96

Decision:

$$t_{\text{calculated}} > t_{\text{critical}} \Rightarrow \text{Reject } H_0$$

5. Visualization

1. **Boxplot of Sleep Quality by Activity Level**
 - Shows that Active individuals have higher median sleep quality
 2. **Bar plot**
 - The bar plot visually confirms the **t-test result**, showing that **Active individuals have higher mean sleep quality than Inactive individuals**
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6. Conclusion

At the 5% significance level, there is **strong evidence** that:

Physical activity significantly affects sleep quality.

Active individuals tend to have **higher sleep quality** compared to inactive individuals.

This confirms H1 and highlights the importance of regular physical activity for better sleep.